Scheme and answer key for

QP CODE: 25900554

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCPHY200 - PRINCIPLES OF MECHANICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

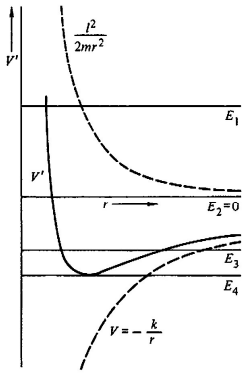
Part A: Short Answer Type Questions

Answer **any 10** questions. Each question carries **2 marks**

1	**Write the equation of motion for a body falling through air under the action of the force of gravity**	
	$$m\frac{dv}{dt} = mg - kv\ (for\ low\ velocity)$$ $$Or$$ $$m\frac{dv}{dt} = mg - Dv^2 (for\ high\ velocity)$$ where m - mass, g - acceleration due to gravity, v - velocity, and $k\ or\ D$ is a constant that depends on the object's shape, the air density, and the drag coefficient. The drag force due to air should be mentioned in the answer.	Any one equation 1 Mark Terms explanation 1 Mark Without drag force equation and terms 1.5 Mark
2	**How is gravitational mass different from inertial mass?**	
	m_G = gravitational mass = the property by which a body exerts a gravitational force on other bodies, and Newton's gravitational law would be expressed as $\vec{F} = -G\left(\frac{m_G M_G}{r^2}\right)\hat{r}$ and m_I = inertial mass = the property by which a body resists a change in its state of motion, so that Newton's second law would be expressed as $\vec{F} = m_I\vec{a}$.	Statements or Equations with terms explained 2 marks
3	**Consider an object of mass $m = 1$ kg that is initially at rest and at the origin. Determine the velocity of the object when it reaches $x = 3\ m$ if it is acted upon by a force $F = 4x$.**	
	$$\frac{1}{2}\ mv^2 = F(x)dx,$$ $$Velocity, v = \sqrt{\frac{2}{m}\int_{x_0}^{x} F(x)dx} = \sqrt{\frac{2}{1}\int_0^3 4xdx} = 6\text{m/s}$$ Or $F(x) = m\frac{dv}{dt} = m\frac{dx}{dt}\frac{dv}{dx} = mv\frac{dv}{dx} =$ kx , k = 4 on integrating v = 6m/s	Equation only 1 mark Equation with proper substitution and correct result 2 marks

4	**How does the gravitational force between two bodies change if the distance between them is doubled?**	
	The gravitational force between two objects, $F = G\frac{m_1 m_2}{r^2}$. If the distance between two bodies is doubled, the gravitational force between them becomes one-fourth of the original force.	Equation only 1 mark; proper explanation 2 Marks
5	**If masses are present in a region, the gravitational potential cannot be zero, whereas the gravitational field can be zero. Comment on this statement**	
	The gravitational field is the rate of change of gravitational potential (V) with distance, $F = -\nabla V$, If F = 0, $-\nabla V = 0$. Ie, the potential is constant over that region, but not necessarily zero. If the gravitational potential is zero, its rate of change must also be zero, which means the gravitational field must be zero.	Equation/ statement- 1 mark Explanation 1 mark
6	**What are the differences between gravitational force and gravitational field?**	
	Gravitational force / **Gravitational field**	Any one difference 2 marks
	The actual force of attraction between two objects with mass. / A region of space around a mass where a gravitational force is exerted on other masses.	
	Depends on both the masses of the two objects and the distance between them, described by Newton's Law of Universal Gravitation / The force per unit mass at a specific point, calculated by dividing the gravitational force by the mass of the object in the field	
	It is the result of a mass interacting with a gravitational field / It is the cause of the force on a test mass placed within it.	
7	**Why is the motion of a pendulum becomes simple harmonic motion for small angles only?**	
	The restoring force is proportional to $\sin(\theta)$. At small angles $\sin(\theta) \approx \theta$ which makes the restoring force directly proportional to the displacement, a condition required for SHM. For larger angles, this approximation is no longer valid, and the motion is no longer purely simple harmonic.	Proportionality 1 mark Explanation 1 mark
8	**Write the general solution for the overdamped case of a damped oscillator.**	
	$x(t) = C_1 e^{-\gamma_1 t} + C_2 e^{-\gamma_2 t}$, C_1 and C_2 are arbitrary constants determined by initial conditions, γ_1 and γ_2 are two distinct, real, and negative roots derived from the characteristic equation of the damped harmonic oscillator's differential equation.	Solution with terms explained 2 marks. Equation/ definition of damped oscillator only 1 mark
9	**A stone tied to a string is rotated in a horizontal circle of radius 0.75m with angular velocity $\vec{\omega} = 5$ rad/sec. Calculate the centrifugal force on the stone of mass 0.25 kg**	
	Centrifugal force $F_c = mr\omega^2$ $F_c = 0.25 \times 0.75 \times 5^2 = 4.6875$ N	Equation 1 mark solution with unit 1 mark

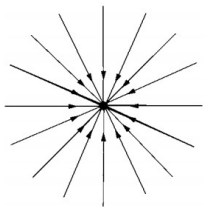
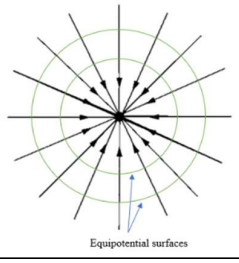
10	**State and explain conservation of angular momentum with an example**	
	The rate of change of the angular momentum of a particle is equal to the net torque exerted on it by external forces. If there is no net external torque acting on a particle, the time rate of change of its angular momentum is zero. Example: Any	Statement 1 mark Explanation 1 Mark Example 1 mark. Any two- 2 Marks
11	**Explain the significance of conservation of angular momentum in the case of Earth-Moon system.**	
	The Moon's gravity raises two tidal bulges on Earth, and because Earth rotates faster than the Moon orbits, the bulge nearest the Moon is dragged slightly ahead. This creates a gravitational torque that slows down Earth's rotation, decreasing its rotational angular momentum. At the same time, this torque transfers angular momentum to the Moon, causing it to move gradually farther away from Earth and increase its orbital angular momentum. Although Earth's rotation slows, the total angular momentum of the Earth–Moon system remains constant, and over billions of years, this process will continue until Earth's day becomes equal to the Moon's orbital period, making one day as long as one month.	Conservation of angular momentum only 1 mark
12	**Is Coriolis force affected the weather on the earth? Justify**	
	Yes, by influencing the direction of large-scale air movements. **Northern Hemisphere:** Moving air deflects to the right. **Southern Hemisphere:** Moving air deflects to the left Coriolis force is an apparent force resulting from the planet's rotation, and it plays a vital role in shaping major weather patterns, from global winds to tropical cyclones.	Yes, 1 Mark Justification 1 mark. If definition only 1 mark
13	**What is the relation between a central force and the corresponding potential energy**?	
	$\vec{F}(r) = -\nabla U$. Where U – Potential energy	Equation 1 Mark, explanation to terms 1 mark. Definition of central force/potential energy 1 mark
14	**What is effective potential energy in central force motion? Illustrate it with energy diagram for an attractive inverse square law field.**	
	Effective potential energy combines the actual potential energy ($\frac{-GMm}{r}$) with the angular part of the kinetic energy ($\frac{l^2}{2mr^2}$) into a single, one-dimensional potential.	Proper Illustration with terms explained- 2 marks Definition- 1 mark Definition with proper

	 **FIGURE 3.3** The equivalent one-dimensional potential for attractive inverse-square law of force.	explanation-2 marks

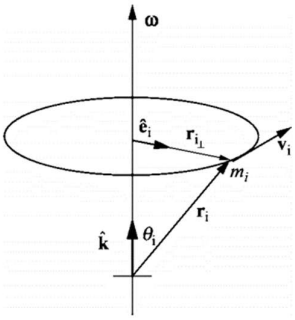
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Part B : Short Essay Type Questions

Answer **any 6** questions. Each question carries **5 marks**

15	**A particle of mass 2 kg is acted upon by three forces (in Newtons), $\boldsymbol{F_1 = 2\hat{\imath}}$, $\mathbf{F_2 = 6\hat{\jmath} + 4\hat{k}}$ and $\mathbf{F_3 = 10\hat{k}}$. Find the acceleration of the particle.**	
	$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$ $\vec{F}_{net} = 2\hat{\imath} + (6\hat{\imath} + 4\hat{\jmath}) + 10\hat{k}$ $\vec{F}_{net} = 8\hat{\imath} + 4\hat{\jmath} + 10\hat{k}$ According to newtons second la w of motion $\vec{a} = \frac{\vec{F}_{net}}{m}$ $\vec{a} = \frac{8\hat{\imath} + 4\hat{\jmath} + 10\hat{k}}{2}$ $4\hat{\imath} + 2\hat{\jmath} + 5\hat{k}$ (in m/s^2) OR $\sqrt{45}$ m/s^2 ≈ 6.71 m/s^2	If the unit is not mentioned, reduce ½ mark
16	**a) Draw the gravitational field lines due to an isolated point mass.** **b) Sketch the corresponding equipotential surfaces.** **c) Explain the nature of these field lines and equipotential surfaces.**	
	(a)  (b)  The field lines are radially inward straight lines, indicating an attractive force whose strength decreases with the square of the distance. The equipotential surfaces—representing constant potential—are three-dimensional concentric spheres centred at the source, and they are always drawn perpendicular to the field lines; their spacing decreases near the centre.	a) 1.5 Marks b) 1.5 Marks c) nature of field lines -1 mark Nature of equipotential surface -1 mark
17	**A particle performs simple harmonic motion with amplitude 5 cm and time period 2 s. At what displacement is its speed half of the maximum speed?**	
	The maximum speed $v_{max} = \omega A$ Where $\omega = \frac{2\pi}{T}$ (no relevance in this problem), T= 2 s, A = 5 cm speed v= $\omega\sqrt{A^2 - x^2}$	Equations with substitution & answer with unit- 5 marks

	when v = ½ v_{max}; $\frac{1}{2}\omega A = \omega\sqrt{A^2 - x^2}$ $x^2 = \frac{3A^2}{4}$; $x = \pm \frac{\sqrt{3}\times 5}{2}$ cm or $x \approx \pm 4.33\ cm$ (sign is not important)	Each step-1 mark
18	**What is a damped harmonic oscillator? Write the equation for the damping term. Obtain the differential equation of damped harmonic oscillator.**	
	A damped harmonic oscillator is a system that experiences a periodic, oscillating force, but is also subjected to an external, non-conservative force that is proportional to the velocity and opposes the motion. This dissipative force causes the system's mechanical energy and the amplitude of oscillations to decrease gradually over time until the system eventually comes to rest. The equation for the damping force, $\vec{F}_{damping} = -b\vec{v} = -b\frac{dx}{dt}$ Where *b* is the damping coefficient $\vec{v}$ is the velocity of the oscillator The forces acting on a damped oscillator of mass m are: Restoring force $\vec{F}_{restoring} = -kx$ and Damping force $\vec{F}_{damping} = -b\vec{v} = -b\frac{dx}{dt}$ So the net force acting on the oscillator is $\vec{F}_{net} = m\vec{a}$ $\vec{F}_{restoring} + \vec{F}_{damping} = m\frac{d^2x}{dt^2}$ $-kx - b\frac{dx}{dt} = m\frac{d^2x}{dt^2}$ Or $m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$	Definition 2 mark Equation of the damping term 1 mark Obtaining differential equation 2 Marks
19	**Obtain the particular solution of a forced damped harmonic oscillator.**	
	The equation of motion for a forced damped harmonic oscillator (FDHO) is a non-homogeneous second-order linear differential equation $m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = F_0 sin\ \omega_d t$ Define a complex function and substitute in the above equation Solve to obtain the particular solution	Definition of FDHO 1 mark. DE – 1 mark Assumption & substitution of complex function 1 mark Obtaining particular solution 2 marks
20	**Derive an expression for angular momentum of a rigid body about a fixed axis**	
	The linear velocity of particle I is related to the angular velocity ω by $\mathbf{v}_i = \boldsymbol{\omega} \times \boldsymbol{r}_{i\perp}$, If we place the origin at the center of the circular path of some particle, say m_i, then it will not be at the center of the path of another particle, say mj, that is displaced somewhat along the axis of rotation. The angular momentum of particle I can be expressed in terms of the angular velocity	Correct derivation 5 Marks (Definition of rigid body/angular

	$\mathbf{l}_i = \mathbf{r}_i \times \mathbf{p}_i = m_i(\mathbf{r}_i \times \mathbf{v}_i) = m_i\mathbf{r}_i \times (\boldsymbol{\omega} \times \boldsymbol{r}_i)$. Using the BAC – CAB rule for the vector triple product, $\mathbf{l}_i = m_i\left[r_i^2\boldsymbol{\omega} - (\mathbf{r}_i \cdot \boldsymbol{\omega})\,\mathbf{r}_i\right]$  We choose ω *is along the z axis*, then $\mathbf{r}_i = r_i \cos\theta_i\hat{\mathbf{k}} + r_i \sin\theta_i\hat{\mathbf{e}}_i$, where θ_i is the angle between r_i and ω, and $\widehat{e_i}$ is a unit vector in the plane of the circular path and pointing (instantaneously) towards m_i, that is, along $r_i \perp$. Using this expression for r_i , $\mathbf{l}_i = m_i r_i^2\omega \sin^2\theta_i\hat{\mathbf{k}} - m_i r_i^2\omega \sin\theta_i \cos\theta_i\hat{\mathbf{e}}_i$. In a symmetrical body, for every particle I, there is an identical particle j diametrically opposite to it at the same distance from the axis of symmetry. The jth particle is identical to the ith, except for the direction of its angular momentum. The components of l_i and I_j parallel to the rotation axis add together, but the components perpendicular to the rotation axis cancel each other. The total angular momentum of a symmetrical body yields $\mathbf{L} = \sum_i m_i r_i^2\omega \sin^2\theta_i\hat{\mathbf{k}}$. The quantity $r_i \sin\theta_i$ is the perpendicular distance from the axis of rotation to the ith particle. If this distance is denoted by $r_i \perp$. We can write $\mathbf{L} = \left(\sum_i m_i r_{i\perp}^2\right)\omega\hat{\mathbf{k}}$.	momentum 1 mark)
21	**Derive the relation $L = \mathbf{L_c} + \mathbf{L'}$ where $\mathbf{L_c}$ is angular momentum of the center of mass and $\mathbf{L'}$ is angular momentum relative to the center of mass.**	

	The total angular momentum of a system of particles is $$\mathbf{L} = \sum_i \mathbf{r}_i \times m_i \dot{\mathbf{r}}_i.$$ The position vector of particle i can be expressed as $$\mathbf{r}_i = \mathbf{r}_c + \mathbf{r}_i',$$ where $\mathbf{r}_c$ is the vector to the center of mass and $\mathbf{r}_i'$ is the position of particle i with respect to the center of mass (see Figure 5.13). Using this relationship, the angular momentum is $$\begin{aligned}\mathbf{L} &= \sum_i m_i \left(\mathbf{r}_c + \mathbf{r}_i'\right) \times \left(\dot{\mathbf{r}}_c + \dot{\mathbf{r}}_i'\right) \\ &= \sum_i m_i \mathbf{r}_c \times \left(\dot{\mathbf{r}}_c + \dot{\mathbf{r}}_i'\right) + \sum_i m_i \mathbf{r}_i' \times \left(\dot{\mathbf{r}}_c + \dot{\mathbf{r}}_i'\right) \\ &= \sum_i m_i \mathbf{r}_c \times \dot{\mathbf{r}}_c + \sum_i m_i \mathbf{r}_c \times \dot{\mathbf{r}}_i' + \sum_i m_i \mathbf{r}_i' \times \dot{\mathbf{r}}_c + \sum_i m_i \mathbf{r}_i' \times \dot{\mathbf{r}}_i'.\end{aligned}$$ The quantities $\mathbf{r}_c$ and $\dot{\mathbf{r}}_c$ do not depend on i and can be taken out of the summations. Also note that $\sum_i m_i = M$ (the total mass). Consequently, $$\mathbf{L} = M\left(\mathbf{r}_c \times \dot{\mathbf{r}}_c\right) + \mathbf{r}_c \times \left(\sum_i m_i \dot{\mathbf{r}}_i'\right) - \dot{\mathbf{r}}_c \times \left(\sum_i m_i \mathbf{r}_i'\right) + \sum_i m_i \left(\mathbf{r}_i' \times \dot{\mathbf{r}}_i'\right).$$ But the two middle terms are zero because $\sum_i m_i \mathbf{r}_i' = 0$ and $\sum_i m_i \dot{\mathbf{r}}_i' = 0$ by the definition of the center of mass. (The first of these is proportional to the position of the center of mass with respect to itself and the second is proportional to the velocity of the center of mass with respect to itself.) The expression for $\mathbf{L}$ then reduces to $$\mathbf{L} = M\left(\mathbf{r}_c \times \dot{\mathbf{r}}_c\right) + \sum_i m_i \left(\mathbf{r}_i' \times \dot{\mathbf{r}}_i'\right),$$ which can be written $$\mathbf{L} = \mathbf{L}_c + \mathbf{L}'.$$ Here $\mathbf{L}_c = M\left(\mathbf{r}_c \times \dot{\mathbf{r}}_c\right)$ is the angular momentum of a particle of mass M located at the center of mass and moving with the center of mass, and $\mathbf{L}' = \sum_i m_i \mathbf{r}_i' \times \dot{\mathbf{r}}_i'$ is the sum of the angular momenta of all the particles relative to the center of mass. That is what we set out to prove.	Correct derivation 5 Marks (Definition of centre of mass/angular momentum 1 mark)
22	**State Kepler's laws of planetary motion**	
	First law(Law of orbits): All planets move in elliptical orbits with the Sun situated at one of the foci of the ellipse. Second law (law of areas): The line that joins any planet to the Sun sweeps equal areas in equal intervals of time. Or Areal velocity is a constant Third law (Law of periods): The square of the time period (T) of revolution of a planet is proportional to the cube of the semi-major axis (a) of the ellipse traced out by the planet ($T^2 \alpha a^3$)	Statement (or mathematical expressions with terms explained) of First/second law 1.5 Marks Third law 2 Marks All 5 Marks

[5 x6=30]